

TAP AGAINST ME

Comprehensive Game Design, Product, Content, Economy, Multiplayer, and Technical Bible

“Every Tap Against Me Takes You Further.”

Document	Value
Version	1.0 - Consolidated design bible
Date	July 12, 2026
Project	Tap Against Me (TAM)
Studio / Umbrella	Don't Watch Media
Primary Platform	Android-first, portrait-oriented mobile; broader mobile support planned
Current Engine	Godot
Status	Active development; competitive core and online infrastructure in progress
Source	Consolidated from project design conversations and implementation decisions through July 12, 2026

Canonical intent

This document separates locked decisions from proposed, deferred, and archived concepts. It is intentionally comprehensive. Where earlier ideas conflict with later direction, the later decision is treated as canonical and the earlier idea is preserved as historical context rather than silently discarded.

0. How to Use This Bible

A single source of truth for design, development, content, economy, presentation, and product strategy.

This bible is meant to let a developer, artist, composer, designer, or future collaborator understand Tap Against Me without reconstructing the project from scattered conversations. It records not only what the game is, but why its systems exist, how they interlock, which boundaries must not be crossed, and what should be postponed until the core experience has proven itself.

Status	Meaning
LOCKED	A confirmed rule or strong constraint that implementation should preserve.
PLANNED	A committed direction that may still require tuning or production sequencing.
PROPOSED	A supported idea that should remain configurable until validated.
DEFERRED	Useful future scope that should not block the current milestone.
ARCHIVED	An earlier direction that is not part of the current canonical design unless deliberately revived.

The design should be implemented as data-driven systems wherever practical. Exact prices, probabilities, thresholds, and timing values should remain configurable. The identity, fairness rules, mode boundaries, and reward ownership rules are more important than any single tuning number.

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1. Executive Vision

The compact statement of what TAM is, what it promises, and why it can deserve a player's time.

Tap Against Me is an Android-first, portrait-oriented, mobile-friendly 2D arcade survival game in which players fall from the sky and tap to remain airborne. Every tap applies upward force and can also influence horizontal direction. Gravity grows more demanding over time. Touching the ground ends the competitive round. The game can be played alone, privately with friends, or through public and ranked online competition.

The immediate fantasy is simple: resist the fall. The deeper fantasy is mastery under increasing pressure. A new player understands the objective in seconds, but sustained survival requires rhythm, positioning, item judgment, risk tolerance, and composure. The game should create repeated “one more try” moments without requiring a lengthy tutorial or a large time commitment.

TAM also supports a relaxed sister experience called Infinity Taps. Infinity Taps removes failure, competition, and pressure. It turns the same tapping language into open-ended fidgeting, exploration, discovery, music, and cosmetic expression. The competitive game asks, “How long can you survive?” Infinity Taps asks, “Where can one more tap take you?”

Core promise

Tap to stay airborne. Adapt as gravity intensifies. Collect what helps, survive what hurts, and outlast yourself or everyone else.

Dimension	TAM Competitive Core	Infinity Taps
Emotional goal	Tension, mastery, rivalry, close calls	Relaxation, fidgeting, curiosity, discovery
Failure	Ground contact ends the round	No loss state
Session shape	Short, repeatable rounds	Continuous, persistent play
Primary metric	Survival, wins, rank, personal best	Lifetime taps, travel, discoveries
Social model	Solo, private, public, ranked, bots	Noncompetitive; shared universe and profiles
Economy	Hero Tokens and rewards feed the cosmetic shop	No coin pickups; 1 Hero Coin per 1,000 valid taps
Exclusive value	Ranked and achievement recognition	Mode-exclusive discoveries and cosmetics that cannot be purchased

2. Identity, Brand, and Player Fantasy

How the title, tone, visual language, and “me” concept reinforce the experience.

The official slogan is “Every Tap Against Me Takes You Further.” It works across both sides of the product: competitively, every tap delays elimination and pushes the player beyond a previous limit; in Infinity Taps, every tap literally advances travel, total taps, and discovery.

The word “Me” is intentionally flexible. It can represent the opponent on the other side of the match, the gravity trying to pull the player down, the player's previous best, the game itself, or the part of the player that wants to stop. This ambiguity strengthens the title rather than weakening it. The game does not need a single literal character named “Me.” The title is a challenge: tap against the force, the rival, and your own limit.

Brand element	Direction
Title	Tap Against Me; TAM as the compact project and community shorthand.
Slogan	Every Tap Against Me Takes You Further.

Brand element	Direction
Competitive tone	Energetic, mischievous, readable, tense without becoming hostile or militaristic.
Infinity tone	Playful, curious, calming, alive, and quietly surprising.
Visual identity	Retro arcade pixel art with a modern, clean mobile presentation; characters remain readable at small portrait-screen scale.
Character tone	Cute, strange, expressive, collectible, and capable of playful dark humor.
Commercial tone	Cosmetic-first, transparent, fair, and explicitly non-pay-to-win.

The game must not look or feel like a generic reskin of a one-button mobile prototype. Its distinction should come from the combination of tactile tap physics, strong character identity, reactive cosmetics, ranked competition with human-like bots, and the unexpectedly large, calm universe of Infinity Taps.

3. Design Pillars and Non-Negotiable Boundaries

The rules that should guide decisions when individual feature requests compete.

- **Immediate comprehension.** A player should understand the main rule—tap to avoid the ground—within a few seconds.
- **Depth through control, not complexity.** The challenge comes from timing, momentum, gravity, positioning, and item decisions rather than a dense button layout.
- **Every tap should feel good.** Animation, sound, haptics, particles, squash-and-stretch, and responsive physics are not polish added at the end; they are the product.
- **Fair competition.** Money may buy expression and convenience around advertising, never power, survivability, improved physics, better odds, or ranked advantage.
- **Short sessions, long identity.** Rounds are compact, but accounts, cosmetics, badges, achievements, ranks, and seasons give players a long-term reason to care.
- **Population resilience.** The game must remain playable when there are few human players. Bots, solo play, private rooms, and eventually ghosts protect the experience.
- **Infinity without pressure.** Infinity Taps must remain a no-loss, noncompetitive fidget-and-exploration experience rather than slowly turning into another survival mode.
- **Rare means memorable.** Aliens, secret characters, lightning, golems, snowmen, and rare cosmetics should feel discovered—not scheduled so often that they become routine.
- **Data-driven growth.** Content definitions, prices, achievement targets, bot traits, seasons, and effects should be extendable without rewriting the core game.

Hard prohibition: no pay-to-win

Do not sell stronger characters, reduced gravity, additional ranked lives, superior fruit odds, faster item pickup, larger collision benefits, score multipliers, ranked revives, or any other purchase that changes the probability of winning.

4. Competitive Core Gameplay

The canonical airborne survival loop and the physics language shared by solo and multiplayer competition.

4.1 Round Start and Objective

A round begins with the player character entering from the sky or already descending within a bounded 2D arena. The player is always under downward gravity. Tapping creates an upward impulse. The objective is to remain in the playable area and avoid touching the ground for as long as possible. In multiplayer, the last surviving player wins unless a specific alternate ruleset is active.

1. Countdown and character introduction.
2. Players begin falling under the same approved ruleset.
3. Taps apply upward force and directional influence.
4. Gravity progressively increases according to the match curve.
5. Items spawn and create movement decisions.
6. Ground contact eliminates the player.
7. The round resolves when the solo player falls or a multiplayer winner remains.
8. Results, rewards, rank changes, achievements, and rematch options are shown.

4.2 Tap Input and Direction

The tap must be processed immediately. On mobile, perceived latency is fatal to trust: players need to believe the character moved because of their exact tap. The simplest competitive mapping is that every valid tap applies an upward impulse while the horizontal position of the tap influences left or right movement. This keeps the one-touch identity while allowing meaningful positioning.

Tap region	Competitive behavior
Left of character / left screen region	Upward impulse with leftward influence.
Right of character / right screen region	Upward impulse with rightward influence.
Near character / central region	Mostly vertical correction with limited horizontal drift.
Rapid taps	More frequent impulses, constrained by physics and reasonable input validation rather than an artificial low cap.

The system should allow skilled rhythm and emergency tapping, but it must not reward machine-like spam above all other behaviors. Momentum, gravity growth, item placement, and arena boundaries should make controlled tapping more effective than blindly tapping at maximum speed.

4.3 Gravity Curve

Gravity is the main difficulty escalator. It should begin forgiving enough that a first-time player can recover, then rise in clearly felt stages. The player should sense increasing demand without the game abruptly changing rules. A configurable curve should control base gravity, increase rate, stage transitions, maximum gravity behavior, and any sudden-death escalation.

- **Opening phase.** Teaches the rhythm and allows item positioning.
- **Pressure phase.** Requires more consistent taps and sharper corrections.
- **High-gravity phase.** Close calls become common; fruit decisions matter more.
- **Maximum-gravity phase.** Survival becomes a mastery test and supports achievements such as Terminal Velocity.

The gravity curve should be synchronized and authoritative in online matches. It must not be calculated independently from untrusted client clocks.

4.4 Loss, Elimination, and Spectating

Touching the ground ends the player's competitive run. The game should communicate the exact moment cleanly with a brief impact, elimination effect, and result state. In multiplayer, eliminated players may spectate, react with limited emotes, or leave without affecting the match. They must not be able to create hazards or obscure active players.

The retry loop must be fast. A solo player should be able to restart almost immediately. Public matches should offer a clear return-to-queue action, while private matches should prioritize rematch flow.

5. Items, Pickups, and Risk

The three foundational pickup types and the role each plays in movement decisions.

Item	Function	Design purpose	Rules
Green Apple	Increases jump force.	Creates rescue opportunities and encourages directional movement.	Helpful power-up; cumulative behavior and caps remain configurable.
Purple Poison Fruit	Increases the gravity affecting the player.	Introduces danger, temptation, achievements, and comeback stories.	Debuff must be clearly telegraphed; cannot be sold or used as paid sabotage.
Golden Hero Token	Persistent currency pickup used in the cosmetic shop.	Connects active play to long-term ownership and identity.	Spawn rate must respect economy; online collection must be validated.

Items should spawn in positions that create decisions rather than simply appearing on the player. A valuable token near the ground is meaningful only when the player voluntarily risks descending. A Green Apple above the player can offer relief but require overcorrection. A Poison Fruit should be avoidable enough that collecting it feels like a mistake, a deliberate challenge, or a chaotic consequence—not an unavoidable punishment.

The item system also feeds achievements: consuming hundreds of fruits over time, surviving after poison, collecting equal fruit totals, or winning while poisoned. This turns the same small content set into multiple long-term goals without changing the competitive rules.

6. Game Modes and Product Structure

How players enter the same core physics through different levels of pressure, competition, and continuity.

Mode	Status	Purpose
Solo Survival	LOCKED / MVP	Personal bests, practice, achievements, fast offline play.
Local Play	PLANNED	Accessible same-location play; exact shared-device or local-network format remains to be finalized.
Private Online Match	LOCKED / IN PROGRESS	Room-code competition with friends and controlled testing.
Public Casual Matchmaking	LOCKED / IN PROGRESS	Low-friction online play with bot backfill when needed.
Ranked Matchmaking	PLANNED	Glicko-based seasonal progression, tiers, rewards, and competitive identity.
Ghost Competition	DEFERRED	Asynchronous competition against recorded human runs.
Infinity Taps	LOCKED CONCEPT; PRODUCT BOUNDARY UNDER REVIEW	No-loss fidgeting, exploration, world discovery, and exclusive rewards.

Current product-boundary decision

Infinity Taps began as a local mode inside TAM, but its scale and emotional identity increasingly resemble a sister product. The safest roadmap is to share characters, accounts, cosmetics, backend, and art systems while preventing Infinity Taps from delaying the competitive TAM launch. A future standalone title may be branded “Infinity Taps: A Tap Against Me Adventure.”

6.1 Solo Survival

Solo survival is both a complete game mode and the training ground for every competitive system. It must work offline, restart instantly, record personal bests, update eligible achievements, and remain worthwhile even when online services are unavailable. Solo is not a lesser mode; it is the most reliable test of whether the core mechanic is intrinsically enjoyable.

6.2 Private Online Matches

Private matches use room codes so players can find one another even when public discovery, friends lists, or matchmaking are incomplete. Private rooms should allow a host to select supported rules, invite players, fill open spaces with bots when desired, and rematch quickly. Ranked rating does not change in private matches.

6.3 Public Casual and Ranked

Public casual play prioritizes fast starts and low stakes. Ranked play prioritizes match quality and competitive integrity. Both should prefer real players, expand the search range gradually, and then backfill with appropriately modeled bots rather than allowing queues to appear abandoned.

7. Online Multiplayer, Matchmaking, and Bots

Population resilience, fair synchronization, and human-like competition without perfect tapping machines.

7.1 Match Authority

The online game should be server-authoritative for match state that affects outcomes. The server validates tap timing, applies approved physics, advances gravity, resolves item collection, determines elimination, and records the final result. Clients render immediate local feedback and may predict movement for responsiveness, but they do not have final authority over victory, coins, or ranked rating.

- Use the same deterministic or tightly controlled simulation rules for humans and bots.
- Validate tap timestamps and reject duplicates or impossible sustained rates.
- Use server-approved item spawn seeds or server-generated spawns.
- Record match IDs and client result IDs to prevent duplicate reward submissions.
- Treat disconnects consistently; short grace windows may permit reconnection, but the server decides the final state.

7.2 Bot Backfill

Bots are a structural feature, not an embarrassment to hide. A small multiplayer game cannot depend on a large simultaneous population at every skill level. Public matchmaking should prefer humans, wait for a configurable period, broaden the acceptable range, and then insert bots so the player can actually play.

Bots must use the same physics, gravity, arena boundaries, tap limits, item rules, character collision, and match conditions as players. Difficulty comes from decision quality, timing consistency, risk tolerance, target selection, and error patterns—not private advantages.

Bot behavior dimension	Human-like implementation
Tap timing	Reaction delay, variable rhythm, missed windows, panic bursts, and recovery attempts.
Movement	Imperfect horizontal correction, momentum overshoot, and occasional poor positioning.
Item response	A configurable recognition delay so bots do not instantly snap toward newly spawned items.
Risk	Different personalities: cautious, greedy, poison-tolerant, token-

Bot behavior dimension	Human-like implementation
	focused, or stability-focused.
Skill	Derived from survival distribution, consistency, recovery quality, and decision accuracy.
Mistakes	Intentional but believable errors; never arbitrary immediate suicide solely to hand the player a win.
Identity	Can use legitimate characters, skins, badges, and names but should not impersonate a known real player.

A player who averages approximately 15 seconds should not routinely face a player or bot averaging 72 seconds. A player averaging 63 seconds can reasonably face one averaging 72 seconds because the matchup remains aspirational rather than impossible. Survival time is not the rating system by itself, but it is an important secondary match-quality signal.

7.3 Ghost Opponents

Ghosts are a deferred asynchronous option. A ghost replays a validated real run using recorded tap inputs or state samples. Ghosts can fill time-shifted challenges, friend comparisons, and low-population events without pretending that a live person is currently connected. They should be labeled clearly as recorded opponents.

8. Ranked Ladder, Glicko-2, and Seasons

A visible Hearthstone-inspired ladder built on a more reliable underlying rating model.

Ranked competition uses Glicko-2 rather than a simple raw Elo counter. Glicko-2 tracks rating, rating deviation, and volatility, allowing the system to represent uncertainty for new or inactive players. Initial implementation values may begin around rating 1500, rating deviation 350, and volatility 0.06, but all values should remain configuration-driven and tested against real distributions.

Visible tier	Divisions	Notes
Bronze	V to I	Entry tier; generous protection and educational bot support.
Silver	V to I	Developing consistency and basic item control.
Gold	V to I	Competent survival and adaptation.
Platinum	V to I	Strong control under pressure.
Diamond	V to I	High-skill competitive population.
Master	Rating-based	Top players; divisions may be replaced by rating position.
Grandmaster	Rating / leaderboard	Highest visible distinction and seasonal prestige.

- **Placement.** Five placement matches establish an initial visible position while Glicko uncertainty remains high.
- **Matchmaking weighting.** A starting proposal is 70% Glicko match quality, 20% survival-time similarity, and 10% connection quality.
- **Expected match quality.** Prefer matchups in which either side has roughly a 40–60% modeled chance of winning.
- **Visible progression.** Use stars or equivalent progress between divisions for understandable feedback while preserving Glicko underneath.
- **Win streaks.** Hearthstone-inspired bonuses may accelerate lower-tier progression, but should taper or disappear in high ranks.

- **Rank floors.** Protect major milestones so experimentation and short bad streaks do not erase all visible progress.
- **Bot policy.** Bots can support lower and mid tiers; their frequency should diminish at high ranks, where human competitive integrity matters most.
- **Season length.** Ninety days, aligned to a three-month content season.
- **Reset.** Use soft rating compression rather than deleting skill. Visible rank resets downward while prior performance still informs matchmaking.

Season-end rewards are cosmetic and recognition-focused: badges, profile frames, titles, trails, tap effects, skins, or Hero Coins. Rank rewards never grant next-season power. Ranked seasonal rewards and the paid Season Pass may share a theme, but buying the pass must not grant rank rewards.

9. Accounts, Profiles, Social Identity, and Badges

The persistent layer that makes a short arcade round matter beyond the moment.

Accounts support Google sign-in through the project's authentication provider, persistent player records, cross-device access, public profiles, owned cosmetics, badge loadouts, achievements, rank, seasonal history, and online presence. A player record should include at minimum a stable player ID, provider subject ID, email where permitted, display name, creation time, last online time, and online status.

9.1 Badge Loadout

Badges are permanent collectibles earned from achievements, seasons, ranked performance, exploration, or supporter recognition. They are not consumed when equipped. The player can equip one primary badge and up to three showcase badges.

Badge placement	Use
Primary badge	Appears beside the username, in multiplayer introductions, match results, and leaderboard rows where space permits.
Showcase badges	Up to three additional badges shown on the full profile.
Collection	Displays every unlocked badge, locked silhouettes where appropriate, source, tier, and description.
Validation	The server filters the loadout to badges the account actually owns.

Tiered achievement badges may evolve from Bronze to Silver, Gold, and Mythic. The highest version is the default profile representation, while earlier versions remain visible in collection history.

10. Progression, Hero Coins, and Retention Loops

Long-term goals that support the arcade core without replacing it.

Hero Coins are the primary cosmetic currency. Players earn them through Golden Hero Token pickups, achievements, challenges, seasonal rewards, selected exploration milestones, and Infinity Taps at a rate of one coin per 1,000 valid taps. Hero Coins are spent on characters, skins, particle effects, tap effects, and profile customization.

Progression must create goals without turning the game into a job. The strongest loop is layered: immediate survival, round rewards, daily and weekly challenges, permanent achievements, badge collection, character/cosmetic ownership, ranked seasons, and rare Infinity Taps discoveries.

Time horizon	Player goal
Seconds	Recover from a fall, reach an item, survive the next gravity increase.
Minutes	Set a personal best, win a match, finish a daily objective.

Time horizon	Player goal
Days	Save for a cosmetic, complete weekly challenges, build a badge loadout.
Months	Climb a 90-day ranked season, complete the Season Pass, collect exclusive cosmetics.
Long term	Master characters, complete achievements, discover rare Infinity content, build a recognizable profile identity.

11. Achievements, Challenges, and Permanent Recognition

A centralized, event-driven system that rewards skill, persistence, experimentation, collection, and secrets.

Achievements are permanent one-time objectives. Daily and weekly challenges rotate. Badges are permanent profile collectibles. Rewards are granted automatically and exactly once; the player should not lose a reward because a claim button was missed.

Achievement	Requirement	Reward
Life of Poisonous Fruit	Consume 500 Poison Fruits.	10 Coins + Poison Fruit Badge
Anti-Gravity	Consume 500 Green Apples.	10 Coins + Golden Apple Badge
Fruit Salad	Collect a Green Apple and Poison Fruit within five seconds.	5 Coins + Fruit Salad Badge
Toxic Appetite	Consume five Poison Fruits in one round.	10 Coins + Toxic Slime Badge
Poison-Proof	Survive 30 seconds after consuming three Poison Fruits in one round.	15 Coins + Poison Shield Badge
Apple a Day	Collect at least one Green Apple on seven consecutive days.	15 Coins + Apple Calendar Badge
Orchard Raider	Collect 25 Green Apples in one round.	20 Coins + Apple Basket Badge
Clean Eating	Survive two minutes without touching a Poison Fruit.	10 Coins + Pure Apple Badge
No Boost Needed	Survive 90 seconds without collecting a Green Apple.	15 Coins + Natural Flyer Badge
Bad Decisions	Consume three Poison Fruits before the first Green Apple of the round.	5 Coins + Questionable Fruit Badge
First Flight	Survive 30 seconds.	5 Coins + Tiny Wings Badge
Learning to Fly	Survive one minute.	10 Coins + Silver Wings Badge
Gravity Defier	Survive two minutes.	20 Coins + Gravity Badge
Sky Eternal	Survive five minutes.	50 Coins + Eternal Wings Badge
Terminal Velocity	Survive 20 seconds after maximum gravity is reached.	25 Coins + Falling Star Badge
Close Call	Recover after entering the configured close-call zone near the ground.	5 Coins + Scraped Paw Badge
Living on the Edge	Trigger five close-call recoveries in one round.	15 Coins + Razor Edge Badge
Perfect Control	Survive one minute without touching top or side boundaries.	15 Coins + Balance Badge
Untouchable	Survive one minute without collecting an item.	15 Coins + Empty Sky Badge
One More Tap	Remain near the ground for ten continuous seconds and survive.	10 Coins + Last Tap Badge

Achievement	Requirement	Reward
Coin Collector	Hold 500 Coins before the achievement reward is granted.	25 Coins + Coin Collector Badge
Golden Hoard	Hold 2,500 Coins before the achievement reward is granted.	50 Coins + Golden Vault Badge
Token Hunter	Collect 1,000 gameplay Coins cumulatively.	25 Coins + Token Hunter Badge
Big Spender	Spend 500 Coins cumulatively.	15 Coins + Shopping Bag Badge
Shop Regular	Purchase ten distinct shop items.	20 Coins + Merchant Badge
Fully Dressed	Equip a skin, particle effect, tap effect, and badge.	10 Coins + Wardrobe Badge
Golden Opportunity	Collect three Golden Hero Tokens in one round.	15 Coins + Golden Hero Badge
Shiny Distraction	Lose within two seconds of collecting a Golden Hero Token.	5 Coins + Dropped Coin Badge
First Victory	Win the first online match.	10 Coins + First Crown Badge
Last One Floating	Win ten online matches.	25 Coins + Survivor Crown Badge
Sky Champion	Win 100 online matches.	100 Coins + Champion Wings Badge
Photo Finish	Win when the final opponent is eliminated within one second of your own near-elimination window.	20 Coins + Photo Finish Badge
Flawless Victory	Win without consuming a Poison Fruit.	15 Coins + Perfect Crown Badge
Poisoned Champion	Win after consuming at least three Poison Fruits.	25 Coins + Toxic Crown Badge
Comeback Flight	Become the lowest active player and still win.	15 Coins + Comeback Badge
Crowded Skies	Win a full-capacity match.	25 Coins + Crowded Sky Badge
Rivalry	Defeat the same player in three completed matches.	10 Coins + Rival Badge
Friendly Competition	Complete ten private matches with friends.	15 Coins + Friends Badge
Getting Started	Complete five achievements.	10 Coins + Beginner Badge
Badge Collector	Unlock ten badges.	20 Coins + Badge Case Badge
Achievement Hunter	Complete 25 achievements.	50 Coins + Achievement Hunter Badge
Completionist	Complete every enabled standard non-seasonal achievement.	Exclusive Skin + Completionist Badge
Dedicated Tapper	Complete 100 rounds.	20 Coins + Tap Counter Badge
Tap Veteran	Complete 1,000 rounds.	75 Coins + Veteran Badge
Persistent	Immediately restart after 25 separate losses.	15 Coins + Never Quit Badge
Daily Flyer	Play at least one completed round on seven consecutive days.	20 Coins + Weekly Wings Badge

11.1 Secret Achievements

Secret achievements appear as hidden entries until completed. The game may reveal that a secret exists, but not its exact requirement.

Secret	Hidden condition	Reward
Was That Worth It?	Lose while moving toward a Poison Fruit.	5 Coins + Regret Badge
Floor Inspector	Lose within three seconds of round start.	Floor Badge

Secret	Hidden condition	Reward
Maximum Poison	Consume seven Poison Fruits in one round.	25 Coins + Purple Skull Badge
Perfectly Balanced	End a round with equal Apple and Poison Fruit counts.	10 Coins + Balance Fruit Badge
Velvet's Nine Lives	Trigger nine close calls in one round while using Velvet Hex.	25 Coins + Nine Lives Badge
Too Greedy	Lose within one second of collecting a Golden Hero Token.	Greedy Hero Badge

11.2 Tiered Achievement Support

Tier	Example Poison Fruit target	Example reward
Bronze	25	3 Coins
Silver	100	5 Coins
Gold	500	10 Coins + Poison Fruit Badge
Mythic	2,500	50 Coins + Animated Poison Badge

The data model should support tiers even if the MVP launches with only selected tiered achievements. The same pattern can apply to Apples, survival time, rounds, victories, coins, close calls, and Infinity Taps.

11.3 Daily and Weekly Challenges

Cadence	Examples	Reward direction
Daily	Collect 10 Apples; survive 60 seconds; play five rounds; trigger two close calls; collect three Tokens; finish a no-fruit round.	2–5 Coins each; optional completion bonus.
Weekly	Survive 20 cumulative minutes; win ten online matches; collect 100 Apples; consume 25 Poison Fruits; earn 100 gameplay Coins; complete five dailies.	10–25 Coins, badge progress, or temporary/seasonal cosmetics.

Challenge resets use UTC or trusted server timestamps. Expired challenges stop progressing. Competitive challenge progress should come from confirmed match results, not a client-only claim.

12. Cosmetic Economy and Shop

Expression is the primary reason to spend; performance remains equal.

The shop sells presentation, identity, and collection. It does not sell survival. The loadout supports a character, character skin, character particle effect, tap effect, and profile badge. Additional future slots may include profile frame, entrance animation, elimination effect, victory pose, emote, arena theme, or fruit reskin.

Shop category	Examples
Characters and skins	Velvet Hex, themed variants, seasonal forms, colorways, alternate materials.
Tap effects	Pulsing rings, rain, snow, magma, fire, leaves, portals, runes, bubbles, flowers.
Character particles	Slime droplets, shadow smoke, sparkles, fire, snow, bats, cosmic dust, glitch fragments.

Shop category	Examples
Profile	Badges, frames, titles, username treatments, seasonal history.
Presentation	Spawn animations, landing effects, elimination effects, victory poses, emotes.

Every cosmetic should be previewable before purchase. Tap effects need an interactive tap test area. Particle effects need idle, jump, fall, and landing previews. Character skins must preserve readable collision and should not provide a visibility advantage.

13. Monetization and Advertising

A cosmetic-first commercial model with clear prices and no permanent power purchases.

Product	Price	Contents and rules
Hero Coin Starter Pack	\$0.99	One-time high-value starter amount sufficient to begin buying cosmetics; example starting point: 150 Hero Coins.
30-Day Ad-Free Pass	\$2.99	Removes forced/interstitial ads for 30 days. Optional rewarded ads remain available. Not permanent.
Season Pass	\$4.99	Three-month free and premium tracks with unique seasonal cosmetics obtainable only during that season.
Supporter Bundle	\$9.99	Character or themed skin bundle, matching character particle effect, matching tap effect, supporter badge, and 60 days ad-free; optional small Hero Coin bonus.
Individual cosmetics	Variable	Hero Coin prices based on complexity; no gameplay advantage.

The 30-day ad-free entitlement begins at purchase and displays its expiration date. The Supporter Bundle grants 60 days. Entitlements should use trusted timestamps and platform purchase validation where possible. Cosmetics remain permanently owned; ad-free time expires.

The Season Pass lasts approximately 90 days and contains free and premium tracks. Purchasing the pass unlocks access to the premium track; it does not instantly grant every item. Players earn progress through play and challenges. Once the season ends, unfinished exclusive rewards become unavailable, while already earned rewards remain permanently owned.

Ad placement	Rule
During a live round	Never.
After every loss	Avoid; it destroys the fast-retry loop.
Forced interstitial	At most after several solo rounds with a real-time cooldown; absent for active ad-free players.
Rewarded ad	Optional only: solo revival, post-round currency multiplier, challenge reroll, or temporary cosmetic preview.
Ranked flow	Avoid forced interruptions between matches; queue continuity matters.

14. Tap Effects

Cosmetics that appear at the exact input point and make each tap expressive.

Tap effects trigger at the valid tap location. They can create temporary environmental reactions, but they never change jump strength, gravity, movement, collection range, score, coins, or competitive visibility. In multiplayer, the owner may see the full effect while other players see a reduced or disabled version.

Effect	Price	Behavior
Purple Pulsing Rings	100 Coins	Expanding purple ring per tap.
Custom Ring Color	50 Coins	Unlocks configurable color; store hue/saturation/brightness/RGB/opacity settings.
RGB Pulsing Rings	300 Coins	Random, smooth rainbow, or custom palette cycling.
Rain	200 Coins	Localized rain shower; 0.001% lightning strike that briefly scorches the ground.
Snow	200 Coins	Snow accumulates, stacks, melts, and may produce an Evil Snowman from a large pile.
Magma / Lava	200 Coins	Molten accumulation flows and cools; a large pool may produce an Obsidian Golem.
Fire	75 Coins	Affordable flame, sparks, embers, and short smoke.
Autumn Leaves	200 Coins	Mixed fall leaves accumulate; landing in a pile scatters leaves across the screen.

Backlog effect	Suggested price	Distinctive behavior
Bubble Pop	100	Interactive floating bubbles; rare giant bubble.
Pixel Lightning	175	Arcade electricity and chain-static buildup.
Starfall	200	Falling stars and temporary constellation patterns.
Galaxy Portal	250	Small cosmic portal; rare alien peek-through.
Ghost Hands	175	Ghostly paws or hands wave, clap, or form an audience.
Slime Splash	150	Temporary goo and tiny slime appearances.
Ink Blot	125	Dripping random silhouettes and symbols.
Paint Burst	100–300	Single color, custom palette, or rainbow environment paint.
Flower Bloom	200	Temporary garden; butterflies or rare giant flower.
Mushroom Growth	200	Clusters, spores, and rare walking mushroom.
Crystal Growth	225	Growing crystal cluster and rare guardian.
Mini Earthquake	175	Cracks, stones, dust, and a harmless rock creature.
Water Splash	150	Puddles, reflections, and rare frog.
Ocean Wave	225	Foam, wave motion, fish, bottle, or tiny boat.
Sand Burst	175	Temporary piles and buried discoveries.
Candy Explosion	200	Bouncing candy and rare candy creature.

Backlog effect	Suggested price	Distinctive behavior
Confetti Cannon	100	Simple celebratory burst.
Musical Notes	150 per set	Tap-position pitch and instrument themes.
Retro Arcade Hit	125	POW, TAP, stars, score text, and CRT motifs.
Glitch	250	Localized RGB separation and digital fragments.
Shadow Creatures	225	Small living shadows that hide or run.
Tiny Footprints	75 / 250 bundle	Paws, demon prints, alien tracks, and other trails.
Magic Runes	225	Connected glyphs and temporary summoning circle.
Black Hole	300	Pulls cosmetic particles only.
Cloud Puff	125	Temporary clouds and rare shapes.
Fireworks	200–350	Colors, shapes, and coordinated displays.
Miniature Meteor	225	Crater, smoke, and rare cosmic object.
Hearts	75	Pixel, gothic, broken, slime, or rainbow hearts.
Bat Swarm	175	Small bats and rare larger companion.
Tiny Companion Reaction	250–400	Brief animated miniature companion per tap.

Not every effect should contain a rare summon. Approximately one-third can have accumulation or discovery behavior; the rest should remain immediate and clean so special events retain meaning.

15. Character Particle Effects

Persistent cosmetics attached to player movement rather than input location.

Character particle effects follow the player during idle, jump, falling, movement, swimming, or landing. A premium effect may include multiple components, but the player equips one character particle package at a time. Attachment anchors should include center, back, feet, head, and tail.

Particle	Price	Behavior
Purple Slime Droplets	100	Droplets separate and splash; bubbles underwater.
Shadow Smoke	150	Dark curls with purple highlights and idle wisps.
Golden Sparkles	150	Golden trail and jump burst.
Poison Bubbles	125	Purple/green bubbles and rare skull symbols.
Pixel Stars	100	Simple retro star trail.
Fire Trail	175	Flames, sparks, embers; steam-like fallback underwater.
Blue Ghost Fire	200	Orbiting spirit flames.
Snowflake Trail	150	Snow, frost mist, and landing puff.
Autumn Leaf Trail	150	Multicolor rotating leaves.
Flower Petals	150/theme	Cherry, rose, sunflower, or fantasy petals.
Bubble Trail	125	Floating translucent bubbles.
Ocean Foam	150	Water droplets on land, bubbles underwater.
Lightning Static	200	Localized electric arcs with safe flash limits.
Rainbow Trail	300	Smooth, pastel, or arcade rainbow.

Particle	Price	Behavior
RGB Pixel Trail	300	Random, cycling, or custom-palette squares.
Glitch Fragments	250	Pixel breakup and localized RGB offset.
Cosmic Dust	250	Stars, tiny planets, and orbiting idle particles.
Mini Comets	225	Small comets intensified during fast falling.
Black Hole Orbit	300	Cosmetic particles orbit a tiny void.
Tiny Bats	200	Small bats scatter; one may idle nearby.
Ghost Companions	250	Tiny expressive ghosts follow and wave.
Tiny Slimes	250	Brief miniature slime followers.
Cat Paw Prints	100	Surface paw prints or translucent air symbols.
Demon Footprints	125	Purple burning prints and spade motifs.
Hearts	75–125	Standard, gothic, broken, pixel, or rainbow.
Musical Notes	150	Visual notes synchronized to tap sound themes.
Confetti	125	Lightweight celebration trail.
Candy Trail	175	Bouncing sweets.
Crystal Shards	175	Colored fragments and sparkle.
Mushroom Spores	175	Glowing spores and occasional tiny mushroom.
Fireflies	175	Independent orbiting lights.
Cloud Trail	150	Cloud puffs and launch-like ascent.
Rain Cloud Companion	250	Delayed cloud above the player.
Halo Light	225	Soft rays and apex halo.
Dark Aura	200	Purple-black outline pulse and rising particles.
Crown Sparkles	250 or reward	Ranked/achievement prestige treatment.
Retro Score Numbers	150	Cosmetic +10, NICE, WOW, or TAP text.
CRT Afterimage	225	Cyan/magenta retro afterimages.
Matrix Code	200	Fictional falling symbols.
Floating Runes	225	Orbiting glyphs and jump circle.
Moon Dust	125	Gray-purple dust and landing puff.
Alien Spores	200	Green-purple particles and alien-eye shapes.

Premium set	Contents	Price
Velvet Hex Shadow Set	Purple slime trail, shadow aura, rune jump burst, slime landing splash.	400 Coins
Cosmic Traveler Set	Cosmic dust, orbiting stars, comet jump, black-hole landing.	500 Coins
Frozen Spirit Set	Snow trail, frost aura, crystal jump, snow landing.	400 Coins
Infernal Set	Fire trail, ember aura, flame jump, magma landing.	400 Coins
Retro Arcade Set	RGB pixels, CRT afterimage, star burst, TAP landing text.	500 Coins

16. Characters, Skins, and Visual Direction

Collectible personalities that make the physics recognizable and the cosmetics desirable.

Characters are the face of TAM. They must be visually readable at small mobile scale, expressive under squash-and-stretch, and compatible with tap, particle, elimination, and Infinity Taps interactions. Characters should not have paid competitive stat advantages. Any future mechanical differences require equal access, rigorous balance, and a separate design decision.

16.1 Velvet Hex

Velvet Hex is a tuxedo-cat demon mascot concept: black-and-white tuxedo markings, purple horns, purple button-like X eyes, pink inner ears and nose, a small bow tie, tiny fangs, and a devil tail with a purple spade tip. The original lore line is: “A stitched tuxedo fiend that trades cuddles for tiny curses.”

Visual development began with a stitched plush demon and moved toward a slime-bodied interpretation. The current desired sprite direction is deliberately lower-fidelity retro arcade pixel art rather than polished high-resolution illustration. “32-bit” and “90s arcade” are being used as style shorthand: chunky readable pixels, limited palette, strong silhouettes, restrained shading, and animation that communicates material through shape changes rather than smooth rendering.



Velvet Hex early plush-demon concept. The identity is useful; the final production sprite style remains under iteration.



Velvet Hex slime jump-study reference. Treat as visual development, not final production art or final frame alignment.

Velvet Hex requirement	Direction
Identity cues	Tuxedo cat pattern, demon horns, X/button eyes, bow tie, fangs, spade tail.
Material direction	Slime or goo-like body that squashes, stretches, pools, and rebounds.
Pixel direction	Lower-detail 90s arcade sprite; crisp clusters, limited colors, minimal anti-aliasing.
Animation	Idle wobble, crouch/squash, takeoff stretch, rise, apex wobble, landing splat/recovery.
Variants	Dracula/vampire with red/black/purple-trimmed coat and red horns; beachwear/bikini variant; Toxic, Golden, Frozen, Arcade, Molten, Ghost, Corrupted, Strawberry Slime.
Competitive readability	Variants cannot hide collision, blend unfairly into arenas, or change apparent hit position.

Characters encountered in Infinity Taps may be tapped to reveal a profile card, trial animation, unlock condition, or purchase option when they are ordinary shop content. Infinity Taps-exclusive characters and variants are clearly labeled and cannot be bought.

17. Infinity Taps

A no-loss fidget toy, endless exploration game, cosmetic showcase, and discovery collection in the TAM universe.

Infinity Taps removes the lose factor entirely. The player starts on the ground and taps to increase a permanent total while moving through a persistent world. Coins do not physically spawn. Every 1,000 valid Infinity Taps awards one Hero Coin. The experience is local-first, casual, noncompetitive, and suitable for absent-minded fidgeting while still containing deep optional exploration.

Infinity Taps identity

The player is never told that they failed. Every tap adds permanent progress, every direction may reveal something, and rare discoveries make long casual sessions feel personal.

Rule	Canonical behavior
Failure	None. No lives, health, timers, ground-loss condition, or mandatory reset.
Competition	None. No ranked ladder or PvP inside Infinity Taps.
Input	Tap left/right/above/below to travel contextually; near-character taps produce smaller movement.
Currency	No coin pickups. Award 1 Hero Coin for each 1,000 validated lifetime taps, preserving remainder.
Persistence	Resume at the last saved location with taps, travel, discoveries, altitude, depth, and rewards intact.
Exclusive rewards	Certain skins, tap effects, particles, badges, and variants are earned only here and cannot be purchased.
Purpose	Relaxation, fidgeting, discovery, music, character interaction, and cosmetic expression.

17.1 Movement and World Response

Movement changes by context: bouncing on land, climbing mountains, floating through clouds, drifting in space, diving underwater, sliding on ice, or making longer jumps on low-gravity worlds. These differences are experiential and do not create failure.

The world should acknowledge taps. Water splashes, bushes wiggle, leaves fall, rabbits react, birds startle, frogs jump, fish scatter, and repeated tapping can cause escalating reactions. The design question should repeatedly be: “I wonder what happens if I tap that?”

Interaction layer	Example
First tap	A bush wiggles or a rabbit looks toward the player.
Repeated taps	Leaves fall, an animal moves, or an object changes expression.
Persistent taps	A hidden interaction chain, temporary creature, or alternate reaction appears.
Contextual tap	Water, weather, biome, character, or equipped effect changes the response.

18. Infinity Taps Worlds, Biomes, Depth, and Altitude

A modular connected universe built from reusable scenes, landmarks, transitions, and rare encounters.

Region	Content direction
Basic Field	Starting area; grass, flowers, hills, clouds, butterflies, rabbits, birds.
Farmland	Crops, barns, animals, tractors, scarecrows, harvest events.
Forest	Dense trees, mushrooms, fireflies, owls, hidden paths, ancient trees.
Jungle	Vines, waterfalls, parrots, monkeys, ruins, glowing insects.
Mountains	Slopes, peaks, caves, goats, eagles, summit encounters.
Desert	Dunes, cacti, ruins, mirages, oasis, scarabs.
Volcanic	Lava, ash, obsidian, fire creatures, eruptions, phoenix encounter.
Arctic	Snow, sea ice, polar wildlife, northern lights, ice caves.
Antarctic	Ice shelves, penguins, seals, aurora, deep ice environments.
Ocean Surface	Waves, boats, islands, dolphins, seagulls, floating objects.
Sunlight Zone	Coral, colorful fish, turtles, rays, schools of fish.
Twilight Zone	Dim light, jellyfish, bioluminescence, larger fish.
Midnight Zone	Darkness, anglerfish, sparse lights, strange silhouettes.
Abyssal Zone	Vents, slow movement, unusual organisms, rare wildlife.
Trench / Mariana-inspired Depth	Deepest destination, ancient structures, exclusive characters and effects.
Cloud Layer	Clouds, birds, balloons, rainbows, floating islands.
Storm Layer	Lightning, heavy clouds, strong visual wind, safe relocation events.
Upper Atmosphere	Thin air visuals, curvature, reduced environmental density.
Orbit / Space	Satellites, stations, meteors, comets, astronaut characters.
Moon	Low gravity, moon dust, craters, lunar landmarks.
Alien Worlds	Mars-like desert, ice, slime, crystal, machine, garden, void, candy worlds.

Zones are envisioned at roughly two minutes of traversal under normal engagement. This allows the player to experience most of a biome-specific music section before transitioning. World construction should use authored biome scenes, procedural or reusable segments, landmark scenes, and preloaded transitions rather than one enormous always-loaded map.

18.1 Infinity Movement Boosts

Boost	Behavior
Velocity Increase	Temporarily accelerates travel and creates a stronger sense of momentum.
Tap-and-Hold Jet Engine	Holding produces a jet-like sustained push and a distinct sound/particle state.
Anti-Gravity	Reverses the usual expectation so taps force the player downward.
Extra Bouncy	Produces intentionally absurd jump height and exaggerated squash-and-stretch.

Boosts are playful traversal modifiers, not competitive power. They can be discovered, triggered by biome events, or activated for a limited segment. They should not be sold as coin multipliers or rare-discovery multipliers.

19. Infinity Taps Discovery, Rarity, and Exclusive Ownership

How rare events feel special without becoming exploitative or invisible.

The original rare-effect concept uses approximately a 0.001% per-tap chance, equivalent to one expected success per 100,000 taps. Pure randomness can create extreme unfairness, so the implementation should use eligible checks and a pity system while preserving the emotional rarity.

- Perform a discovery check at intervals, such as every 100 valid taps, rather than every rendered particle.
- A sample equivalent starting rate is 0.1% per 100-tap check.
- Gradually increase the chance after long periods without a rare discovery.
- Reset or reduce pity after a successful discovery.
- Tie some discoveries to biome, depth, altitude, weather, time, character, or landmark conditions.
- Spawn the reward physically and leave it available long enough to notice and collect.

Infinity Taps-exclusive rewards cannot be bought. This is a hard ownership rule. Their value comes from demonstrating that the player spent time in the mode, explored, and encountered something not everyone will see. Store-eligible characters encountered in the world may still present a purchase or unlock path, but an exclusive is never converted into a paid shortcut.

Discovery Journal category	Examples
Biomes and worlds	Field, desert, Moon, trench, alien planet.
Characters	Named heroes, travelers, aliens, secret creatures.
Wildlife	Rabbits, birds, fish, deep-sea animals, cosmic creatures.
Landmarks	Ancient tree, summit, ruins, space station, trench structure.
Events	Rainbow, lightning, meteor, tractor beam, rare weather.
Cosmetics	Exclusive tap effects, particles, skins, badges, profile items.
Records	Highest altitude, greatest depth, total distance, tap milestones.

20. Infinity Taps Milestones and Achievements

Permanent progression for a mode that never declares failure.

Achievement	Requirement	Reward
Just One More Tap	Reach 1,000 Infinity Taps.	Fidget Badge
Tap Enthusiast	Reach 10,000 Infinity Taps.	10 Hero Coins
Endless Motion	Reach 100,000 Infinity Taps.	Endless Tap Badge
Million-Tap Hero	Reach 1,000,000 Infinity Taps.	Exclusive Tap Effect + Badge
World Traveler	Discover ten biomes.	Explorer Badge
Into the Clouds	Reach the cloud layer.	Cloud Badge
Beyond the Sky	Reach space.	Astronaut Badge
One Small Tap	Land on the Moon.	Moon Badge
Going Deep	Reach the Midnight Zone.	Deep Ocean Badge
At the Bottom	Reach the deepest trench.	Abyss Badge
First Contact	Encounter an alien.	Alien Badge
What Are the Odds?	Find an extremely rare cosmetic.	Lucky Tapper Badge
New Friend	Encounter a named character.	Friendship Badge
Around the World	Visit every launch Earth biome.	Globe Badge
Fidget Session	Tap 5,000 times in one session.	Fidget Master Badge

Normal one-coin-per-1,000-taps payouts and achievement rewards must be tracked separately to prevent accidental double-counting. The milestone reward transaction uses a deterministic ID derived from the completed tap threshold.

21. Music, Sound, Haptics, and Audio Footprint

A shared musical identity that changes intensity across menu, competition, and exploration.

21.1 Shared Musical DNA

- Bright electronic percussion.
- Bouncy synth bass.
- Plucked marimba, kalimba, or pizzicato-style melodies.
- Soft arcade tones rather than harsh chiptune overload.
- Organic claps, snaps, wood knocks, bells, whistles, and environmental accents.
- Playful, energetic, slightly mischievous identity that remains replayable over long sessions.

21.2 Main Menu

The main menu uses a calmer and cleaner version of the same identity: soft electronic percussion, warm bouncy bass, gentle plucked melody, subtle arcade tones, and welcoming organic accents. It should invite play without exhausting the player before a round begins.

21.3 Solo and Competitive Rounds

Competitive music is faster, sharper, and more focused than Infinity Taps. Punchy percussion, driving bass, quick plucked melodies, and brief arcade accents should build intensity as gravity rises without becoming stressful or masking input feedback. The mix must leave room for tap sounds, item cues, close-call warnings, and elimination sounds.

21.4 Infinity Taps Adaptive Music

Infinity Taps uses biome-specific music as adaptive layers over a shared core theme. Forest, Snow, Beach or Ocean, Desert, Space, and later biomes retain the same musical DNA while introducing environment-specific instruments, ambience, and melodic variations. Two-minute zones allow the player to experience most of a track or arrangement before transition.

Biome	Possible layer
Forest	Wood percussion, soft whistles, birds, leaves, warm plucks.
Snow / Ice	Bells, soft pads, delicate high plucks, muted percussion.
Beach / Ocean	Light shakers, watery mallets, airy pads, wave ambience.
Desert	Dry hand percussion, sparse plucks, wind, resonant strings.
Space	Floating pads, soft pulses, filtered arcade arpeggios, distant signals.

Reactive moments can temporarily add musical layers when boosts trigger, rare characters appear, the player enters a new depth or altitude, or a long tap streak occurs. These reactions should feel discovered rather than constantly firing.

21.5 Tap Sound and Haptics

Tap sounds need small pitch and sample variation so thousands of inputs do not become irritating. Infinity Taps may support unlockable sound themes such as soft slime, retro arcade, bubble pop, wooden click, gentle chime, or raindrop. Music, ambience, tap sounds, character sounds, cosmetic sounds, and haptics should have independent controls.

21.6 Music Compression and Runtime Guidance

The project currently has roughly eight songs totaling about 27 MB. This is not catastrophic for a mobile game, but it should be managed deliberately. “Lossless with major size reduction” is generally unrealistic for finished music; the practical target is perceptually transparent compression.

- Keep uncompressed master WAV files outside the shipped build and repository package where practical.
- Ship music as Ogg Vorbis, commonly around 96–128 kbps stereo after listening tests; raise only when artifacts are audible.
- Use Godot streaming for longer tracks rather than loading every song into memory.
- Trim silence, create clean loop points, and avoid exporting redundant extended intros/outros.
- Share motifs and layers carefully; adaptive stems can save repetition but can also multiply file count if overused.
- Use mono for small one-shot effects where stereo provides no value; keep music stereo unless a deliberate mono aesthetic is chosen.
- Test on phone speakers, common earbuds, and Bluetooth devices before final bitrate decisions.

22. User Experience, Navigation, and Presentation

The primary flows that connect a fast arcade game to a persistent account and cosmetic collection.

Screen / flow	Core requirements
Main Menu	Play, Infinity Taps, Profile, Achievements, Shop, Season, Settings; immediate access to the last-used mode.
Mode Select	Solo, private room, public casual, ranked, local options, Infinity Taps boundary.
Match HUD	Survival time, relevant item feedback, player state, opponent status; minimal obstruction.
Post-Match	Placement, survival, personal best, coins, achievement progress, rank change, rematch/queue.

Screen / flow	Core requirements
Profile	Character display, primary badge, three showcase badges, rank, season history, achievement summary.
Achievements	Category filters, progress bars, rewards, secret placeholders, completed dates.
Badge Collection	Owned badges, locked silhouettes, equip primary, add/remove showcase.
Shop	Featured, Characters, Skins, Tap Effects, Particles, Profile, Hero Coins; interactive previews.
Season	Free and premium tracks, time remaining, challenge progress, reward preview.
Infinity HUD	Lifetime taps, session taps, location, discovery alerts, 1,000-tap coin progress; no competitive clutter.

Achievement notifications should never cover active competitive play. A compact progress indicator may appear during a round, while the full unlock presentation occurs after the round or as a small non-blocking banner. Infinity Taps notifications also must not interrupt tapping.

Accessibility settings should include particle density, other-player effects, effect opacity, screen shake, flash intensity, lightning flash reduction, sound categories, haptics, and potentially a reduced-motion mode. Cosmetic intensity options should include Full, Reduced, Minimal, and Off.

23. Technical Architecture and Current Implementation Direction

The Godot client, authentication, Cloudflare backend, persistence, and authoritative online plan.

Layer	Current direction
Client	Godot; Android-first portrait mobile build; desktop editor testing and phone deployment.
Authentication	Google sign-in through Supabase; backend verifies Supabase JWT/JWS using supported algorithms and trusted claims.
API / Runtime	Cloudflare Workers.
Relational storage	Cloudflare D1.
Realtime rooms	Durable Objects for lobby/room state and authoritative match coordination.
Backend endpoint	https://tap-against-me.dontwatchmedia.workers.dev
Current deployment status	Health endpoint works; production D1 initialized; schema bootstrap disabled in production.
Core data	Users, achievements, badges, rewards, matches, flags, and related ownership/transaction records.
Room access	Private room codes; public matchmaking and bot backfill.
Client sync	Early basic polling may use approximately 250–500 ms updates; realtime authoritative transport should replace or supplement polling for production match feel.

The current architecture has already moved meaningfully into accounts, profiles, matchmaking, backend routes, database state, cross-device testing, and competitive synchronization. The design should not casually discard that work. Scope control should focus on getting a narrow online loop reliable, using bots and solo play to protect population, rather than restarting the project as an offline-only prototype.

Legacy hand-rolled authentication approaches are retired. The current intended trust boundary is an external identity provider, validated tokens, server-side account records, and server-authoritative ownership and match results. Production schema mutation must remain disabled or strongly protected.

24. System Architecture Inside Godot

Decoupled managers and event-driven content evaluation.

Gameplay objects should emit events; they should not directly unlock achievements, grant purchases, or alter account ownership. Central managers subscribe to events and apply data-driven rules.

Manager / service	Responsibility
EventBus	Signals for round, tap, item, match, cosmetic, shop, and exploration events.
PlayerStats	Lifetime, season, per-round, streak, Infinity, and personal-best statistics.
AchievementManager	Evaluates definitions, progress, compound conditions, and completion.
ChallengeManager	Daily/weekly rotation, expiration, and progress.
BadgeManager	Ownership, loadout validation, primary/showcase behavior.
CosmeticLoadoutManager	Character, skin, particle, tap effect, profile cosmetics.
TapEffectManager	Spawns tap-location cosmetics and accumulation zones.
CharacterParticleManager	Attaches trails, auras, jump, fall, and landing effects.
InfinityTapsManager	Valid taps, movement context, milestone coins, world state.
BiomeManager	Segment loading, transition, palette, ambience, and environment rules.
EncounterManager	Animals, named characters, rare events, and interaction escalation.
DiscoveryManager	Journal, rare checks, pity, exclusive unlocks.
NetworkService	Authentication, rooms, authoritative match communication, reconnect.
SaveService	Versioned local persistence, offline journal, cloud merge, transaction IDs.
Analytics	Privacy-conscious event instrumentation for product decisions.

Definitions should be resources or JSON-like records containing stable IDs, display text, source, price, compatibility, reward, effect scenes, rare-event data, enabled flags, and version. Stable IDs are essential for purchases, seasonal ownership, migrations, and analytics.

25. Persistence, Transactions, Anti-Cheat, and Competitive Integrity

Protecting shared currency and ranked results without punishing legitimate offline play.

- Every currency grant has a unique transaction ID and is idempotent.
- Achievement rewards track `reward_granted` separately from completed.
- Coin-balance achievements evaluate before adding their own reward.
- Infinity coin milestones derive from validated lifetime tap thresholds and previously paid thresholds.
- Online match rewards and rating changes use server-confirmed results.
- Tap validation rejects duplicates and clearly impossible sustained automation patterns while allowing legitimate rapid tapping.
- Offline sessions maintain a signed or checksummed local journal and synchronize summaries when connected.

- Suspicious sessions can be excluded from shared currency confirmation without deleting local noncompetitive exploration progress immediately.
- Purchase receipts are validated through Google Play or Apple and restored after reinstall where applicable.
- Equipped cosmetics and badges are filtered against server-owned IDs.

Infinity Taps is noncompetitive, but its Hero Coins enter the shared economy. This is why tap milestones still require validation. The goal is not invasive anti-cheat; it is to prevent trivial macros or edited saves from destroying the meaning of cosmetics and purchases.

26. Commercial Viability, Validation, and Analytics

How to determine whether players enjoy the game before scaling content and live operations.

TAM has credible small-game commercial potential because the core is understandable, mobile-native, repeatable, cosmetic-friendly, and compatible with short sessions. Its risk is not comprehension; it is depth and retention. Achievements and cosmetics can amplify an enjoyable loop, but they cannot rescue tapping that feels imprecise or rounds that become identical after a few attempts.

The most important emotional proof is whether players voluntarily restart after a loss because they want another run. The most important competitive proof is whether stronger players visibly make better decisions rather than merely tapping faster. The most important Infinity proof is whether players tap even when no reward notification is imminent.

Internal validation metric	Promising target / interpretation
Players completing three rounds	70%+ indicates onboarding and initial feel are working.
Day-one retention	30%+ is a practical early target for an indie test.
Day-seven retention	10%+ suggests the progression layer has traction.
Sessions per active day	2+ indicates a repeatable short-session loop.
Achievement completion	40%+ complete at least one; goals are visible and reachable.
Badge loadout use	15%+ change or equip a badge; profile recognition matters.
Shop open rate	20%+ indicates cosmetic curiosity even before purchase.
Rewarded ad use	10%+ voluntary use can support monetization without forced disruption.
Immediate restart after loss	A core qualitative and quantitative signal of intrinsic fun.
Infinity taps without reward proximity	Evidence that the fidget loop works independently of currency.

These are internal decision targets, not guarantees. The product should instrument round duration, retries, tap cadence, deaths by gravity stage, item pickup decisions, queue time, bot frequency, match quality, cosmetic previews, challenge engagement, Infinity travel, and rare-discovery frustration. Analytics should answer a design question, not merely collect everything possible.

27. Development Roadmap

A staged plan that preserves the long-term vision without allowing it to bury the playable core.

Phase	Primary deliverable	Included scope
Phase 0 - Foundation	Reliable project and backend baseline	Godot project, Android deployment, authentication, D1 schema, protected production bootstrap, health checks, basic profiles.
Phase 1 - Competitive Vertical Slice	One polished round from start to result	Tap physics, gravity curve, ground loss, Apples, Poison Fruit, Hero Tokens, solo survival, fast retry, analytics.
Phase 2 - Online Playable Loop	Two devices can reliably play	Private room codes, authoritative result,

Phase	Primary deliverable	Included scope
		reconnect handling, basic public queue, server validation, cross-device test.
Phase 3 - Population Resilience	Online play remains available	Bot backfill, skill profiles, human-like delay, item response tuning, casual matchmaking, solo/offline fallback.
Phase 4 - Account Progression	Reasons to return and identify	Achievements, badges, profile loadout, Hero Coin wallet, small shop, initial skins/tap/particle effects.
Phase 5 - Ranked Seasons	Competitive long-term structure	Glicko-2, placements, Bronze–Grandmaster, divisions, stars, floors, 90-day season, rewards.
Phase 6 - Monetization and Live Content	Fair commercial launch systems	Starter pack, 30-day ad pass, Season Pass, Supporter Bundle, platform validation, content cadence.
Phase 7 - Infinity Prototype	Prove fidgeting before building a universe	One field, persistent taps, 1 coin/1,000 taps, simple movement, reactive water/bush/tree, one or two animals, one hidden exclusive.
Phase 8 - Infinity Biome Journey	Structured exploration	Two-minute zones, several biomes, boosts, discovery journal, exclusive rewards, adaptive music.
Phase 9 - Infinite Adventure	Large living universe	Ocean depth, sky/space/moon/planets, rare encounters, seasonal worlds, long-term exploration progression.

Scope discipline

Do not build dozens of characters, planets, particles, or ranked tiers before the tap physics, gravity escalation, fast retry, and two-device match are satisfying and reliable. Content scale is valuable only after the foundational loop earns repeated play.

28. Recommended Initial Content Scope

A launch-sized catalog large enough to feel intentional but small enough to polish.

- **Competitive.** Solo survival, private online, public casual with bots, a limited ranked ladder, Apples, Poison Fruit, Hero Tokens, one primary arena, and several gravity stages.
- **Characters.** Four to six distinct characters with Velvet Hex as a flagship; enough skins to demonstrate the system without diluting quality.
- **Cosmetics.** Approximately 8–12 tap effects, 8–12 character particles, several skins, profile badges, and a small curated shop.
- **Progression.** Approximately 15–25 achievements at launch, daily challenges, a badge loadout, and a starter season.
- **Audio.** Main menu theme, competitive track set, tap/item/UI sounds, and an initial Infinity theme if the prototype ships.
- **Infinity prototype.** One alive-feeling field, simple directional movement, persistent taps, milestone coins, a few interactions, and one rare exclusive.

29. Risks and Mitigations

The most likely ways the project can become expensive, unfair, repetitive, or unclear.

Risk	Consequence	Mitigation
Core tapping lacks precision	Players blame the game rather than themselves.	Prioritize input latency, prediction, physics tuning, and device testing before content.

Risk	Consequence	Mitigation
Rounds feel identical	Achievements cannot sustain interest.	Item placement, gravity stages, arena modifiers, and skill expression; add content gradually.
Low concurrent population	Queues feel abandoned.	Solo, private codes, bot backfill, expanding search, ghosts later.
Bots feel perfect or fake	Players distrust results.	Human-like delay, visible error patterns, same physics, item recognition delay, tier limits.
Network complexity overwhelms scope	Development stalls in infrastructure.	Narrow authoritative loop; reuse systems; postpone secondary social features.
Infinity Taps consumes the project	Two products compete for one roadmap.	Shared universe, separate milestones, prototype first, consider sister-app boundary.
Cosmetics obscure play	Competition becomes unreadable.	Local full effects, remote reduced effects, intensity settings, strict visibility rules.
Currency inflation	Shop loses meaning.	Server transactions, configurable earn rates, limited sources, no multipliers sold.
Rare content feels impossible	Players assume it does not exist.	Pity, journal hints, biome conditions, visible collection silhouettes.
Seasonal exclusivity feels predatory	Players resent missed content.	Clear deadlines, fair \$4.99 pass, meaningful free track, no power, preserve owned rewards.
Music size grows uncontrolled	App footprint and memory rise.	Ogg streaming, listening-tested bitrate, trimmed loops, source masters outside build.
Art style drifts	Characters stop feeling from the same game.	Pixel specifications, palette limits, silhouette reviews, shared animation standards.

30. Open Decisions and Decision Register

Questions that remain configurable or unresolved rather than silently assumed.

Decision	Current position	Next validation
Exact competitive horizontal control	Tap position influences direction; exact curve remains tunable.	Phone playtest for precision and one-handed comfort.
Gravity formula and cap	Progressive stages are locked; values are open.	Survival distribution and frustration testing.
Local multiplayer format	Desired, but shared-device versus local network is unresolved.	Prototype both only if low cost; select clearest experience.
Public match player count	Maximum not finalized.	Network, readability, and queue-population testing.
Bot disclosure	Bots are accepted; exact labeling is unresolved.	Trust and UX testing; never imply a live human when legally/ethically misleading.
Rank stars versus pure divisions	Hearthstone-inspired stars are planned over Glicko.	Prototype clarity, rank anxiety, and reset behavior.
Infinity as mode versus separate app	Shared-universe sister-product direction is favored.	Prevent delay to TAM; evaluate prototype retention and app-store strategy.
Character mechanical differences	Cosmetic-only characters are safest current rule.	Do not introduce stats without a separate balance and access design.
Exact Hero Coin pack amount	\$0.99 starter is locked; amount remains economy-dependent.	Price cosmetics, then ensure first purchase feels useful.
Rare discovery odds	0.001% emotional target; interval/pity implementation proposed.	Simulate player distributions and test perceived fairness.
Final Velvet Hex production style	Slime body and lower-detail retro arcade	Create a true grid-aligned 32/64 px production

Decision	Current position	Next validation
	direction.	sprite manually or with pixel cleanup.

31. Archived and Superseded Concepts

Historical ideas retained for context but not part of the current canonical build.

Earlier explorations described TAM as a lighter MMO-inspired tapping game involving collectible Puffs, artifacts, Explore/Journal/Profile sections, passive bonuses, trading, guilds, and alternate match concepts such as First to Zero, Most Taps, or Dual Victory. Those ideas belong to an earlier direction and are not part of the current canonical airborne survival loop unless deliberately reintroduced through a future design review.

The current project identity is the fall-from-the-sky survival game with increasing gravity, fruit risk, Hero Tokens, competitive matchmaking, characters, cosmetics, achievements, and Infinity Taps. Retaining this archived note prevents older documents or code labels from silently pulling the project in two incompatible directions.

32. Glossary

Shared language for design and implementation.

Term	Definition
TAM	Tap Against Me, the competitive airborne survival game and shared universe.
Infinity Taps	No-loss fidgeting and exploration experience in the TAM universe.
Hero Coin	Persistent cosmetic currency.
Golden Hero Token	In-round collectible that contributes to the Hero Coin economy.
Green Apple	Helpful fruit that increases jump force.
Purple Poison Fruit	Debuff fruit that increases gravity.
Tap Effect	Cosmetic spawned at the tap location.
Character Particle	Cosmetic attached to the moving player.
Primary Badge	Single badge displayed prominently beside the username.
Showcase Badge	One of up to three additional profile badges.
Close Call	Recovery after entering a configured danger zone near the ground.
Glicko-2	Underlying ranked rating model tracking rating, uncertainty, and volatility.
Bot Backfill	Insertion of skill-appropriate AI after human matchmaking has waited or expanded.
Ghost	Recorded human run used for asynchronous comparison.
Pity	Gradual rare-discovery chance improvement after repeated failure.
Mode-exclusive	Earned only through a particular mode and unavailable for purchase.

33. Final Product Directive

The shortest possible instruction for every discipline working on the game.

Tap Against Me succeeds only when the act of tapping is satisfying enough to survive without the surrounding systems. Build a precise, readable, escalating airborne survival game first. Make competition available even with a small population through solo play, private rooms, and human-like bots. Let achievements, badges, seasons, characters, and cosmetics transform short rounds into long-term identity without selling power. Develop Infinity Taps as the pressure-free mirror of the competitive game: a living world where every tap is progress, curiosity is rewarded, and exclusive discoveries must be earned rather than bought.

North-star test

Would a player tap again if there were no immediate reward popup—and would they feel that the next tap belonged to them rather than to the game? If the answer is yes, the surrounding TAM universe has something worth building.

A. Appendix - Canonical Snapshot for Development Tickets

A compact checklist for implementation handoffs.

Area	Canonical direction
Platform	Android-first, portrait mobile, Godot.
Competitive core	Fall from sky; tap upward; gravity increases; ground ends round.
Items	Green Apple boosts jump force; Purple Poison Fruit increases gravity; Golden Hero Token feeds cosmetic economy.
Online	Private rooms, public casual, ranked; server authority; bots backfill low population.
Ranked	Glicko-2; Bronze, Silver, Gold, Platinum, Diamond, Master, Grandmaster; 90-day seasons.
Profiles	One primary badge + up to three showcase badges.
Monetization	\$0.99 starter; \$2.99/30-day ad-free; \$4.99 Season Pass; \$9.99 Supporter Bundle/60-day ad-free; no P2W.
Infinity	No loss, no competition, 1 Hero Coin per 1,000 valid taps, exploration and mode-exclusive unpurchasable rewards.
Music	Bright electronic percussion, bouncy synth bass, playful plucked melody, soft arcade and organic accents.
Visual	Retro arcade pixel art; strong silhouettes; Velvet Hex currently slime-bodied with tuxedo-demon identity.
Backend	Cloudflare Workers + D1 + Durable Objects; Supabase/Google authentication; server ownership validation.
Slogan	Every Tap Against Me Takes You Further.